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# Dynamic speed and separation monitoring for collaborative robot applications – Concepts and performance

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ABSTRACT

Speed and separation monitoring (SSM) allows safeguarding the operator in collaborative robot applications by maintaining a certain minimum separation distance during operation. A continuous adaptation of the robot velocity in response to relative operator and robot motion can be employed to improve the efficiency of SSM-type applications.

The present paper presents two approaches for obtaining a robot velocity limit for this adaptation, considering the separation distance as well as the direction of robot motion. Using a collaborative machine tending task as an example, the impact of these approaches on application productivity was assessed in physical trials and compared to that of conventional safeguarding methods, i.e. zone-based supervision and safeguarding by physical barriers. The trials confirmed that the continuous speed adaptation has a notable productivity benefit over the state of industrial practice. Major factors that influence the particular benefit, such as the frequency and timing of operator presence near the robot, have been identified and investigated.

Before these concepts can be applied in industry, machinery safety requirements must be satisfied; at present, this depends particularly on the availability of sufficiently reliable information on the human position, which is not provided by safety sensors today.

## 1. Introduction

An increasing interest in human-robot collaboration (HRC) has been observed throughout industry in recent years, driven primarily by the demand for increased flexibility in production environments. This is a result from the trend towards lower volume and higher mix in production many manufacturers are facing. In addition, the flexibility offered by HRC enables partial automation also in small and medium enterprises, whose typical lot sizes have so far rendered automation uneconomical.

The technical specification ISO/TS 15066:2016 [1] *"Robots and robotic devices – Collaborative robots"* provides a first systematic framework of principles and requirements for the design of HRC applications. It distinguishes between four different types of collaborative robot operation, one of which is *speed and separation monitoring* (SSM), which is the focus of the present paper. In SSM-type applications, the robot maintains a certain minimum distance towards the human during operation; thus any contact with the moving robot is avoided. It can therefore be applied to enable collaboration using standard industrial robots (also ones already present in an existing production environment that shall be adapted for HRC).

Another collaboration principle, *power and force limiting* (PFL), in contrast, allows for direct physical contact between the operator and the robot. However, this typically requires specially designed manipulators, which limit the energy transfer on potential contact by design or control. Furthermore, PFL falls short or demands special provisions for certain types of applications (e.g. high payloads, sharp or pointed tools) where contact cannot be accepted.

The rest of the paper is structured as follows: The remainder of Section 1 provides further background information on SSM and potential realizations. Section 2 discusses the state of the art concerning SSM in industry and related work in academia. How we obtain the permissible robot velocity for the continuous adaptation of robot speed is explained in Section 3. Section 4 introduces the physical prototype setup and the example application that have been used to validate the developed methods. In Section 5, we describe the methodology and test scenarios used to gauge and compare the performance of the system in physical trials. An overview of the obtained results is given in Section 6. Section 7 presents a discussion and detailed examination of particular interesting aspects of the experiments, before addressing limitations of our implementation and suggesting further work. Finally, the conclusions of this work are presented in Section 8.

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### 1.1. Speed and separation monitoring

SSM relies on the continuous supervision of the separation distance $S$ between the operator and the robot. For robot motion to continue, $S$ must not fall below the *protective separation distance* $S_p$ *(SSM criterion)*:

$$S \geq S_p \tag{1}$$

Thereby, the protective separation distance depends on a variety of factors, such as the manipulator's velocity, its stopping distance, the operator's approach velocity, the system response time, measurement uncertainties etc. Properly considering all of these parameters ensures that the manipulator can always come to a stop before the operator is able to reach it.

### 1.2. Types of SSM

ISO/TS 15066 [1] allows for different realizations of SSM within its scope, which could be classified depending on e.g.

- how the protective separation distance is determined, i.e. online, offline based on worst case values; for the entire manipulator or the tool center point (TCP) only;
- how robot motion is affected to mitigate the risk posed to the operator (stop only, speed adaptation, path or task adaptation);
- what kind of information concerning operator exposure is available (access supervision only, zone occupation information, position information).

A classification attempt concerning the latter two aspects is illustrated in Table 1. In the present paper, we assume that information concerning the actual position of the operator is available (which is not provided according to machinery safety requirements by available safety sensors today). In the simplest form, a SSM application can then be realized by permanent supervision of the SSM criterion (1); in case the condition is violated, a stop of robot motion is issued. The protective separation distance $S_p$ is computed either offline, based on worst-case values for the particular application (e.g. maximum robot velocity), or continuously at run-time using the actual values.

Instead of only passively supervising the SSM criterion, one could try to actively influence $S_p$ or $S$ to avoid violation of the condition as long as possible and uphold productivity. One potential option is the continuous adjustment of the robot velocity based on the actual separation distance. Slowing down the robot upon operator approach reduces the protective separation distance, thereby allowing robot operation to continue at a closer distance to the operator (compare [1, clause 5.5.4.2.3 b]). This approach is used in the present work; we refer to it as *Dynamic SSM* or *dSSM* due to the continuous speed adaptation.

Another option is the active modification of the robot path depending on the operator location, such as to maximize the actual separation distance $S$. While there is a plethora of prior work on online collision avoidance that could be applied in this context (see for example [2–4]), their relevance for industrial applications is rather low, since e.g. tight space requirements and the lack of sufficiently comprehensive environment information limit their usefulness. Switching between different programmed paths or tasks, however, is a viable option that is (based on zone occupation information) already applied in industry.

**Table 1**
Classification of SSM-type applications.

| | | Means of influencing robot to mitigate risk: *Stop only* | *Adapt speed* | *Adapt path/task* |
|---|---|---|---|---|
| Information on operator exposure | *Access* | ✓ | × | × |
| | *Zone occupancy* | × | ✓ | ✓ |
| | *Position*[1] | e.g. [5] | dSSM (this paper), [6,7] | e.g. [2–4] |

×: not practical / expedient ✓: state of industrial practice [1]No safety-rated information on operator position available today.

## 2. State of the art

### 2.1. Industrial applications of SSM

Industrial applications using SSM-type safeguarding today are limited by available safety sensor technology, which provides only logical information about operator presence in static zones instead of actual position data. Simple setups can be implemented by means of light curtains and safety contact mats, whereas safety laser scanners [8–10] or camera systems [11] allow for elaborate configurations with numerous and complexly shaped zones – associated with a correspondingly high engineering effort, both up front and upon cell layout changes. Some setups may furthermore take into account the state of the robot in a similar discrete manner (e.g. robot on left / right side). Based on the zone occupation information, the robot motion is typically affected in three discrete levels only:

1. Full speed robot operation while no person is nearby;
2. Optionally, a reduced speed mode is engaged when the operator approaches the robot (using a predefined speed limit). This may prove beneficial in terms of ergonomics and reduces the stopping distance which needs to be considered in the stop zone dimensions.
3. A stop of robot motion is issued once the operator enters an inner zone around the robot, which covers at least the minimum separation distance according to ISO 13855 [12].

In a different approach associated with SSM, the manipulator is equipped with proximity sensors to detect nearby persons and stop prior to an impending collision. Most commercial systems today are based on capacitive sensing technology. An example of this principle is Bosch's *APAS assistant* [13], which uses a Fanuc robot covered with capacitive sensor elements. Their low detection range of about 50 mm requires to limit the robot speed to 0.5 m/s during collaborative operation. Comau's *AURA* robot combines capacitive and piezoelectric sensing layers to detect both proximity as well as contact [14]. Solutions to equip other manipulators with similar sensing capabilities are available or upcoming, e.g. [15,16], furthermore sensors targeted particularly at safeguarding of hazardous robot tools [17,18], or grippers for HRC applications that provide integrated capacitive sensing [19].

### 2.2. Related scientific work

There are various research activities related to SSM, covering areas such as sensor technology, methodologies for determining the minimum distance and permissible robot speed as well as validation of SSM implementations. Apart from the numerous related prior work on collision avoidance topics, some methods and prototypes focused particularly on SSM in the context of the relevant robot safety standards [1,12,20,21] have been presented; all mentioned research prototypes are *functional* implementations only, i.e. they do not fulfill machinery safety requirements as per ISO 13849-1 [22] yet:

Szabo et al. [23] implemented and validated a SSM prototype that uses laser scanners to safeguard a rail-mounted robot. The system keeps track of the human velocity, but does not consider the direction of motion of the robot or the operator. While the robot motion is unaffected except for issuing a stop, Marvel [24] extended this work by introducing a reduced speed state and furthermore proposed a productivity as well as a safety metric to assess the performance of the SSM implementation. A review and critical discussion of the different contributions to the protective separation distance as defined by the technical specification, highlighting the complexity of the SSM problem, is presented in [25]. Another SSM-type concept is presented by Magnanimo et al. [26], who dynamically adapted the supervised zones that safeguard a mobile platform with an articulated robot. Vicentini et al. [5] proposed the computation of the protective separation distance along the entire kinematic chain to obtain a dynamically changing protective separation "envelope" that depends on the robot trajectory.

While the mentioned implementations use a maximum of three discrete speed levels (full speed, possibly one discrete reduced speed level, and stop of robot motion), Lasota et al. [6] implemented a continuous adaptation of robot speed based on a custom function of separation distance; though this approach does not yet account for e.g. the robot's stopping distance systematically. Similarly, the approach presented by Zanchettin et al. [7] employs a continuous adaptation of robot speed based on a simplified distance criterion; the authors furthermore exploit the null space motion capability of a redundant manipulator to maintain sufficient clearance towards the human worker.

As will be shown in the present paper, the potential benefit of SSM-type safeguarding depends on various application-specific parameters. Simulation tools facilitate an individual assessment (concerning e.g. economical factors), thereby supporting the selection of a suitable method to safeguard a particular application, as presented e.g. by Lehment and Boca [27].

Research in related areas which are of major importance for SSM covers topics such as the prediction of the operator's motion and future workspace occupancy [4,28–31], and the estimation of robot braking distances [32–34]. Extensive work on vision-based collision avoidance approaches has been carried out at the University of Bayreuth [3,35–37]. Researchers at Fraunhofer IFF presented approaches based on projections [38] or tactile flooring [39]; a radio-based localization was shown by Rampa, Savazzi et al. [40,41]. Ergonomics in HRC applications are examined e.g. in [42–44].

### 2.3. Contributions

This paper presents two approaches for obtaining a robot speed limit based on the separation distance to nearby users and the robot's direction of motion. The obtained speed limit is used to continuously adjust the robot speed in a SSM-type application. We show the potential performance gain of an example machine tending task with regular operator interaction when using the continuous speed adaptation approaches compared to conventional safeguarding means, i.e. fences with interlocked gates or zone-based supervision via safety sensors, and discuss factors that influence this performance gain.

## 3. Safe robot speed

In the following, we present two options to compute a robot speed limit based on the separation distance between the robot and the operator. The second approach also considers the direction of robot motion in addition.

### 3.1. Preliminary considerations

ISO/TS 15066:2016 [1] defines the protective separation distance $S_p$ as

$$S_p(t_0) = S_h + S_r + S_s + C + Z_d + Z_r \,. \tag{2}$$

By this definition, the problem is limited to one dimension; $S_p$ is expressed as the sum of contributions by robot motion (reaction distance $S_r$, and stopping distance $S_s$), operator motion ($S_h$) and attributes of the sensor systems or measurement principle respectively ($C + Z_d + Z_r$). In this expression $C$ is the intrusion distance according to ISO 13855:2010 [12]. The contributions to $S_p$ by the position uncertainties of the operator and the robot ($Z_d + Z_r$) are neglected here.

Note that $S_p = S_p(v_r^{(i)})$ is a function of the current velocity $v_r^{(i)}$ of a given point ($i$) on the robot manipulator. It is the goal in the following to find an inverse function to (2), such that a *maximum permissible velocity* at this point can be computed as a function of the actual separation distance $S^{(i)}$. By applying the resulting TCP velocity limit to the robot motion it is ensured that the SSM criterion (1) is not violated, and robot motion can continue. For simplicity, the index ($i$) is mostly omitted in the following; nevertheless the derived relations apply for all observed points along the entire manipulator.

#### 3.1.1. Manipulator representation

To obtain a simple and efficient representation for the computations, the manipulator has been approximated by spheres (Fig. 1, compare e.g. [26,28]). CAD-representations in combination with algorithms for computing the minimum distance between two convex polyhedrons (e.g. [45,46]), or approaches especially suited for certain types of sensors [35,47] could be applied instead depending on application requirements (e.g. if better accuracy is necessary, which might be the case in close-proximity SSM applications).

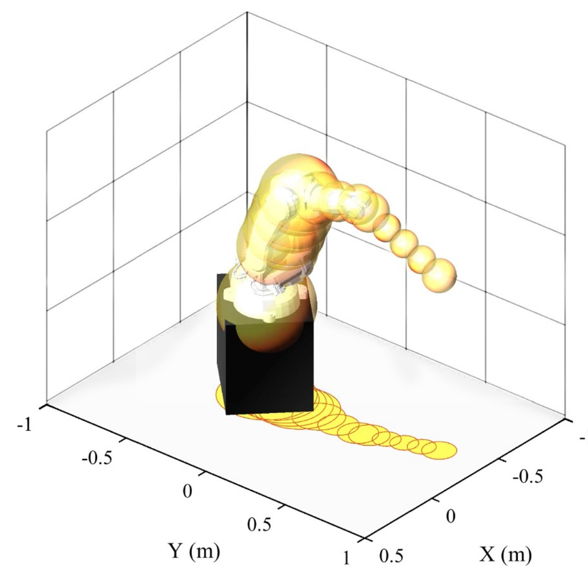



**Fig. 1.** ABB IRB140 manipulator and tool (not shown) approximated by spheres, and their projection on the floor plane.

The sphere positions were obtained via the forward kinematics and interpolation for better approximation of longer links; their respective radii $r_i$ were manually dimensioned such as to encapsulate the manipulator, tool and potential workpiece completely. The number of spheres chosen was a compromise between coverage and computational performance. Specifically for the robot used in this work, the choice of 13 spheres well encapsulated the manipulator geometry with negligible impact on the processing time. Spheres were positioned to cover joints and protrusions as well as evenly distributed along the links in between.

Since we are primarily interested in horizontal relative motion between the operator and the robot, it seems sufficient to limit the considerations to two-dimensional space; furthermore, many sensor systems provide only 2D data concerning the operator position. Consequently, we have chosen to project the sphere approximation onto the horizontal plane, such that the robot arm geometry is represented by a number of circles, as indicated in Fig. 1. These circles are defined by their center positions $\boldsymbol{R}_i$ and radii $r_i$, where $i = 1, \ldots, N_c$ is the index of the circle and $N_c$ is the total number of circles representing the manipulator.

#### 3.1.2. Robot motion

During the response time $T_r$ of the system, we assume constant robot motion with the velocity $v_r$, such that the distance covered by the robot is $S_r = v_r T_r$. After this period, the robot begins to decelerate. The resulting stopping distance $S_s$ depends on a variety of factors, such as the current robot pose and velocity, the inertia of the manipulator including payload, the braking torque etc. Consequently, an accurate computation of $S_s$ requires knowledge about the dynamic properties of the robot. Our approach is based on a simplification instead, by assuming a constant deceleration $a_s > 0$. The value for $a_s$ is taken from product documentation and represents the TCP deceleration for a controlled stop; it is used here as a conservative estimate for the deceleration at each observed point on the manipulator. Then the robot's stopping distance is

$$S_s = \frac{v_r^2}{2a_s}. \tag{3}$$

Future work should include a less conservative estimation, e.g. by use of a verified dynamic model or measurement data.

### 3.1.3. Operator motion

The current operator position, as detected by a sensor system, is assumed to be available as horizontal two-dimensional Cartesian coordinates $\boldsymbol{H}_j$:

$$\boldsymbol{H}_j = \begin{bmatrix} x_{H_j} \\ y_{H_j} \end{bmatrix},$$

where $j$ is an index accounting for the possibility of multiple persons or obstacles detected within the supervised space, which is mostly omitted in the following.

As proposed in the technical specification [1], we assume that the operator motion is directed towards the robot, with an approach velocity of $v_h = 1.6\,\mathrm{m/s}$. The operator's change in location during the system's response time $T_r$ and the stopping time of the robot $T_s$ is given by

$$S_h = v_h (T_r + T_s), \tag{4}$$

where $T_s = v_r/a_s$. Tracking and prediction of operator motion (e.g. velocity, direction, future workspace occupancy) based on sensor measurements could be integrated in the future.

### 3.1.4. Actual separation distance

With the control points representing the robot arm $\boldsymbol{R}_i$ and the operator position $\boldsymbol{H}_j$ as given above, the actual separation distance, i.e. the smallest distance between the manipulator and any sensor detection, is determined by searching for the overall minimum:

$$S = \min_{\forall i,j}(|\boldsymbol{R}_i - \boldsymbol{H}_j| - r_i). \tag{5}$$

The intrusion distance $C$ is considered separately in formula (2) and therefore not subtracted here, which is possible since $C$ is assumed to be equal for all sensor detections $\boldsymbol{H}_j$.

## 3.2. Robot speed limit (dSSM 1D)

Substitution of (2) in the SSM criterion (1) (using the terms introduced above for $S_h$, $S_r$ and $S_s$) yields

$$S \geq S_p(v_r) = v_h (T_r + \frac{v_r}{a_s}) + v_r T_r + \frac{v_r^2}{2a_s} + C. \tag{6}$$

A stop of robot motion must be issued at the instant the actual separation distance $S$ falls below $S_p$, since otherwise the operator would be able to reach the moving manipulator within the system's reaction time $T_r$.

To avoid unwanted stopping, we choose to adapt the manipulator speed based on the actual separation distance $S$ to ensure that $S_p$ fulfills (1). By solving equation (6) for $v_r$ we obtain the updated robot speed limit:

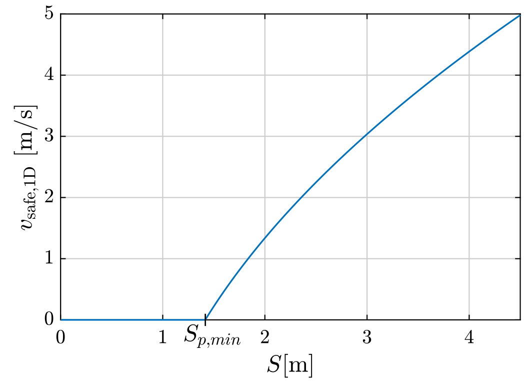



**Fig. 2.** Permissible velocity $v_{\mathrm{safe,1D}}(S)$ (7) as function of separation distance using the test setup parameters.

$$\begin{aligned} v_r &\leq \sqrt{v_h^2 + (a_s T_r)^2 - 2a_s(C - S)} - a_s T_r - v_h \\ &\leq v_{\mathrm{safe,1D}}. \end{aligned} \tag{7}$$

Programmed robot trajectories are updated to conform to this limit. Due to the purely one-dimensional treatment of the problem, we refer to this approach as *dSSM 1D* and the associated speed limit as $v_{\mathrm{safe,1D}}$ in the following.

Fig. 2 shows the permissible robot velocity as a function of the separation distance according to (7). To obtain this curve, the relevant parameters from our physical test setup have been used, which are listed in Table 2. The test setup is introduced in Section 4. Note that the allowable robot speed drops to zero at a positive separation distance of

$$\begin{aligned} S(v_{\mathrm{safe,1D}} = 0) &= v_h T_r + C \\ &= S_{p,min} = S_p(v_r = 0). \end{aligned} \tag{8}$$

as shown in Fig. 2, which corresponds to the distance the operator can cover during the system reaction time.

**Table 2**
Test setup parameters.

| Parameter | Symbol | Value |
|---|---|---|
| Intrusion distance | $C$ | 1.08 m |
| Directed human velocity | $v_h$ | 1.6 m/s |
| System response time | $T_r$ | 210 ms |
| Worst case robot deceleration | $a_s$ | 10 m/s$^2$ |

## 3.3. Considering the direction of robot motion (dSSM 2D)

The approach presented in Section 3.2 is based on the worst-case assumption concerning operator and robot motion (i.e. both moving directly towards each other), and thus fairly conservative. In this section, we present an alternative approach for determining the permissible robot speed that takes into account the actual direction of robot motion. In particular, this approach allows higher velocity for robot motion directed away from the operator in contrast to the previous approach.

The following assumptions have been used:

- The robot continues to move in the current direction during the system reaction time $T_r$. The current direction of motion of a particular point on the manipulator $\boldsymbol{R}_i$ is simply estimated based on the two most recent positions of this particular point:

  $$\boldsymbol{d}_r = \boldsymbol{R}_i(t_0) - \boldsymbol{R}_i(t_{-1}). \tag{9}$$

  where $t_0$ is the current instant and $t_{-1}$ denotes the previous time step. Note that this requires a suitable sampling rate of robot position data, otherwise the direction might be misleading (e.g. in case of circular motion at a low sampling rate, or due to noise for position data sampled at a high rate). Future work could integrate the approach with the robot's motion plan to avoid this potentially erroneous estimation.

- The robot's stopping distance $S_s$ is directed towards the location of the human; as suggested also by ISO/TS 15066 [1]. This covers potential off-path braking motion, e.g. due to use of the mechanical brakes of the manipulator (depending on the selected stopping method).
- The robot maintains the calculated speed and does not accelerate during $T_r$. This assumption does not comply with the technical specification, which states "[...] the system design may use the current speed of the robot, but shall account for the acceleration capability of the robot in the manner that reduces the separation distance the most" [1]. It should be noted, however, that [1] aims at computing the protective separation distance from ongoing robot motion, whereas the goal here is to compute (and apply) a safe velocity limit based on the actual separation distance. Assuming an acceleration beyond this velocity limit would therefore correspond to presupposing a failure in applying this limit to the robot motion.

Fig. 3 illustrates an exemplary scene, examining a single point on the manipulator $\boldsymbol{R}_i$ as well as a sensor detection (i.e. human position) $\boldsymbol{H}_j$ at the time instant $t_0$. Extrapolating the direction of motion from its last two known positions, $\boldsymbol{R}_i$ moves under a certain angle φ in relation to the direction towards the human $\boldsymbol{H}_j$. Using the direction of robot motion $\boldsymbol{d}_r$ from (9), and $\boldsymbol{d}_h$ as the current distance vector:

$$\cos\varphi \ = \frac{\boldsymbol{d}_r \cdot \boldsymbol{d}_h}{|\boldsymbol{d}_r| \cdot |\boldsymbol{d}_h|}, \quad \text{where} \quad \boldsymbol{d}_h \ = \boldsymbol{H}_j(t_0) - \boldsymbol{R}_i(t_0).$$

The aim is to find the permissible robot velocity such that the remaining separation distance between the predicted robot position after the system response time, $\boldsymbol{R}_i(t_0 + T_r)$, and the human position $\boldsymbol{H}_j(t_0)$ is sufficient to cover the estimated human motion (including intrusion distance) as well as the robot's stopping distance (including the robot arm geometry, described by the circle radius $r_i$), i.e.

$$S_h + S_s + C + r_i\,, \tag{10}$$

where $S_s$ and $S_h$ are functions of the robot velocity, whereas $C$ and $r_i$ are constant. This segment can equally be described using the cosine rule:

$$(S_h + S_s + C + r_i)^2 = d_h^2 + (v_r T_r)^2 - 2 d_h v_r T_r \cos\varphi\,. \tag{11}$$

Substitution of the known terms for $S_s$ and $S_h$ from equations (3) and (4), expansion and grouping by equal powers of $v_r$ yields

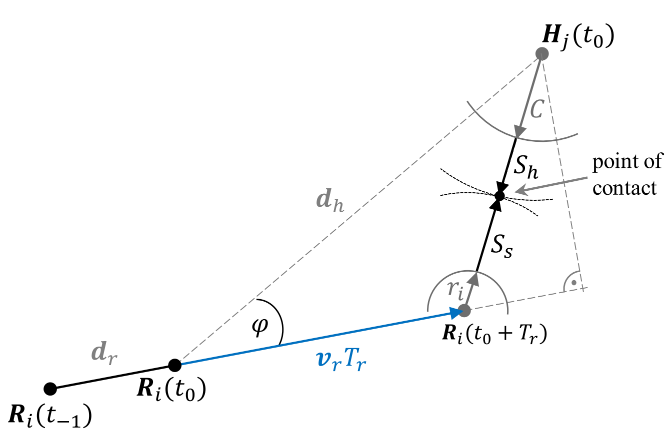



**Fig. 3.** Consideration of the direction of robot motion - schematic example scene, showing one observed point on the manipulator $\boldsymbol{R}_i$ and one point $\boldsymbol{H}_j$ representing the operator position. The direction of robot motion is estimated by linearly extrapolating the vector formed from the two most recent positions. Worst-case assumptions concerning operator motion and the robot's stopping distance are retained.

$$\begin{aligned} 0 \ = \ & \frac{1}{4\,a_s^2} v_r^4 + \frac{v_h}{a_s^2} v_r^3 + \left(\frac{v_h^2}{a_s^2} - T_r^2 + \frac{b}{a_s}\right) v_r^2 \\ & + \left(2 d_h T_r \cos\varphi + \frac{2 v_h b}{a_s}\right) v_r + b^2 - d_h^2\,, \\ & \text{where} \quad b = v_h T_r + C + r_i\,. \end{aligned} \tag{12}$$

Let $v_{sol,k} = f_v(d_h, \cos\varphi),\ \ k = 1 \ldots 4$, describe the solutions of equation (12). Since the separation distance between the respective points $\boldsymbol{R}_i$ and $\boldsymbol{H}_j$, less the circle radius $r_i$, is $S_{ij} = d_h - r_i$, and $S_{ij}$ fulfills the condition $S_{ij} > v_h T_r + C$, that is, $S_{ij}$ is larger than the minimum protective separation distance $S_{p,min}$ (8), then only one real and positive solution to equation (12) exists:

$$\exists\,!k\colon (v_{sol,k} \in \mathbb{R}^+)\,.$$

If $S_{ij} = S_{p,min}$ then one solution is zero, which can be seen by substitution of $d_h = v_h T_r + C + r_i = b$ in (12). However, if the robot is moving away from the human, there may exist a second real and positive solution; i.e. the robot is able to escape the minimum protective separation distance within the reaction time $T_r$. In this case, there is furthermore a small range of distance values $S_{ij}$ very close to but smaller than $S_{p,min}$ for which two real and positive roots can occur. In the following, the larger one of these roots is selected, which allows for a continuous speed adaptation function. For $S_{ij} \ll S_{p,min}$ the polynomial has only negative or complex roots. The larger of the real and positive solutions $v_{sol,k}$, if existent, can thus be used for speed adaptation of the robot; otherwise, the robot is stopped:

$$v_{\mathrm{safe},ij} = \begin{cases} \max\limits_{\forall k}(v_{sol,k} \in \mathbb{R}^+) & \exists\ k\colon (v_{sol,k} \in \mathbb{R}^+)\,, \\ 0 & \text{otherwise}\,. \end{cases} \tag{13}$$

Note that the computation of the permissible velocity is necessary for *each combination* of manipulator points $\boldsymbol{R}_i$ and sensor detections $\boldsymbol{H}_j$, in contrast to the one-dimensional approach presented in Section 3.2, where the velocity computation is required only once, based on the minimum separation distance $S$ which has been found before. Here, with consideration of the robot's direction of motion, the closest object does not necessarily result in the smallest permissible robot velocity.

The permissible robot velocity to apply in the adaptation is then found as the minimum of all computed velocity limits:

$$v_{\mathrm{safe,2D}} = \min_{\forall\, i,j}(v_{\mathrm{safe},ij})\,. \tag{14}$$

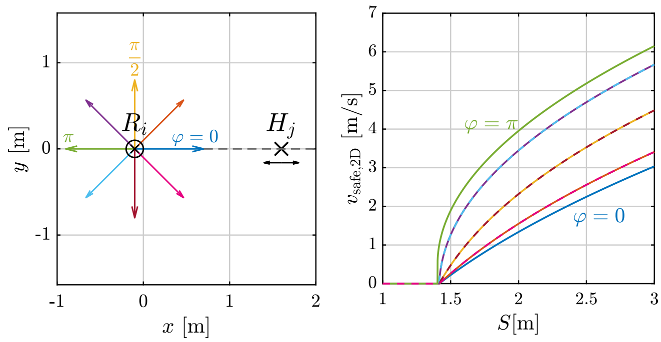
We then adjust the TCP velocity to

$$\tilde{v}_{\mathrm{TCP}} = \min(v_{\mathrm{safe,2D}},\ v_{\mathrm{TCP}}),$$

where $v_{\mathrm{TCP}}$ is the magnitude of the originally programmed velocity for the current trajectory segment. This approach was selected due to the simplicity of implementation in the proprietary ABB robot programming language. A more precise solution would be to reduce the speed of the offending point on the manipulator in a trajectory-consistent manner. By comparison, our approach is more conservative, except near kinematic singularities or for redundant manipulators.

In the following, we use the term *dSSM 2D* to refer to this procedure. Fig. 4 shows an example simulation, in which the robot velocity limit $v_{\mathrm{safe,2D}}$ has been computed as a function of the separation distance for different directions of robot motion with respect to the operator position (represented by the angle φ). The computation is again based on the parameters of our prototype setup (Table 2), which is introduced subsequently in Section 4.

For the case $\varphi = 0$, the manipulator element $\boldsymbol{R}_i$ is moving directly towards the detection $\boldsymbol{H}_j$. Since this direction of motion was presupposed in the one-dimensional approach, the resulting velocity curve is equal to the one shown in Fig. 2. If the robot is moving with an angle $\varphi \neq 0$ with respect to the direction towards the detection instead, the permissible velocity is larger, with the steepest slope in the case of $\varphi = \pi$, i.e. the robot is moving in the opposite direction related to the




**(a)** Point on the manipulator $\boldsymbol{R}_i$ and detection (i.e. operator position) $\boldsymbol{H}_j$ with different directions of robot motion (angle $\varphi$). The separation distance $S$ is varied by moving $H_j$ along the dashed line.

**(b)** Resulting permissible velocity $v_{\text{safe,2D}}$ when varying the separation distance $S$ at fixed angles $\varphi$. For $\varphi = 0$, the robot is moving towards the detection; the resulting velocity limit is equal to the 1-dimensional approach. For $\varphi = \pi$ the robot moves in the opposite direction.

**Fig. 4.** $v_{\text{safe,2D}}(S)$ at different angles of robot motion φ with respect to the direction towards the detection.

detection; note that the velocity limit in this case is more than twice as large.

A scenario in which the robot is passing by in parallel to the detection is illustrated in Fig. 6. The resulting permissible velocity reaches its minimum *before* the minimum of the separation distance (i.e. at $x_{R_i} = 0$), since the smaller approach angle of the robot towards the detection requires a more severe speed reduction despite the larger separation distance. When passing by the detection, the robot can already accelerate slightly as the approach angle increases.

## 4. Prototype implementation

### 4.1. Test setup

The two approaches for obtaining a robot speed limit explained in

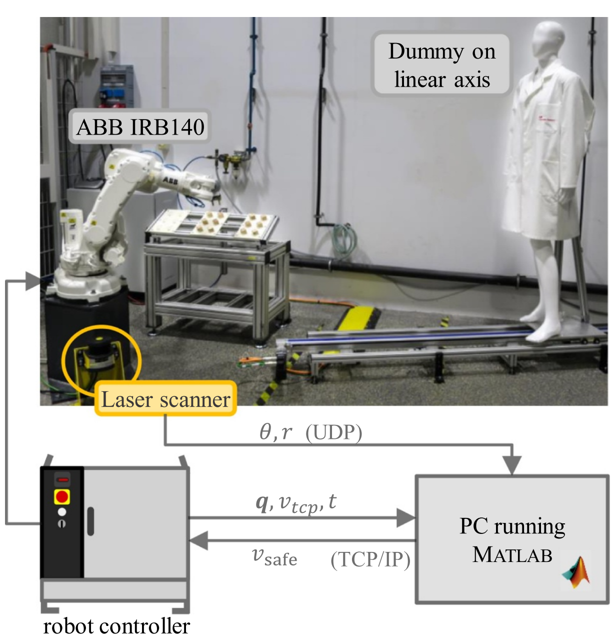
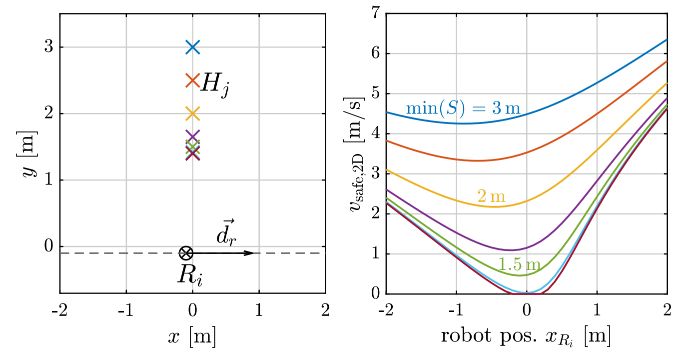



**Fig. 5.** Overview of the prototype setup. The laser scanner transmits its measurement data to an external PC, which computes the robot speed limit and communicates with the robot controller. A dummy mounted on a linear axis can be used to generate repeatable test scenarios.




**(a)** Point on the manipulator $\boldsymbol{R}_i$ passing by (along the dashed gray line) in parallel to a detection $\boldsymbol{H}_j$.

**(b)** Resulting permissible velocity $v_{\text{safe,2D}}$ at different distances.

**Fig. 6.** $v_{\text{safe,2D}}$ when the robot is passing by in parallel to a detection.

Sections 3.2 and 3.3 have been applied to adapt the robot speed in a physical test setup, using an ABB IRB140 industrial robot in combination with a Leuze RSL440 safety laser scanner to obtain the location of approaching persons. Fig. 5 illustrates this setup and component interconnection.

The measurement data from the laser scanner is transmitted to an external PC running Matlab, which performs the velocity computations according to the selected method. The PC communicates with the robot controller in order to obtain the robot position and apply the velocity limit to robot motion. A mannequin mounted on a linear axis was used to generate repeatable "operator" motion in order to compare the performance of both dSSM approaches and of conventional safeguarding means.

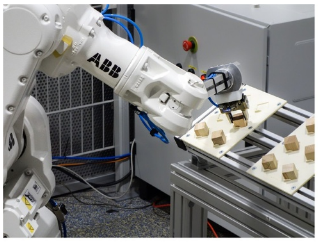
Note that the scan data transmitted by the laser scanner is not safety-rated. To our knowledge, also no other currently available safety sensor provides this capability directly. A workaround could be to use a safety laser scanner with multiple protective zones, providing coarse position resolution at the expense of complex configuration. Further development in the area of safety-rated position determination will be necessary to facilitate productive industrial applications using dSSM.

To ensure safety during testing, we utilize the protective fields of the laser scanner (compare Section 4.3), and require a human approaching the robot to carry a three-position enabling switch; the safety-rated information on zone violation and the status of this switch is used to issue a stop of robot motion if necessary.

### 4.2. Example application

To analyze the performance of the speed adaptation methods, a collaborative machine tending task was implemented in the prototype setup (Fig. 7). The robot feeds an (imaginary) machine with blanks (e.g. a CNC-machine or an injection molding machine) and extracts the finished parts subsequently. The blanks and finished parts are placed in trays, which need to be exchanged by an operator at regular intervals. In one full *application cycle*, the robot handles three trays with six parts each.

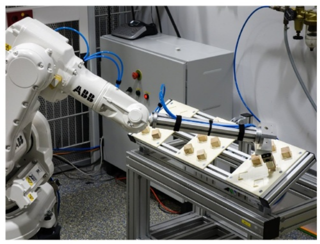
The application can roughly be divided into three phases, namely pick & place phases either on tray or machine side, and transfer motion between those locations. The robot is programmed to move with maximum speed (up to 5000 mm/s) during the transfer motion, and lower speeds of 500 mm/s for motion close to the trays and the machine. The most critical last few centimeters during pick & place operations are performed slowly at 50 mm/s. Concerning the transfer motion, we included the option to switch between three different transfer paths, which are illustrated in Fig. 8:



(**a**) Robot picking blank from tray.

(**b**) Transfer to / from machine location.

(**c**) Inserting blank into machine / extracting finished part.

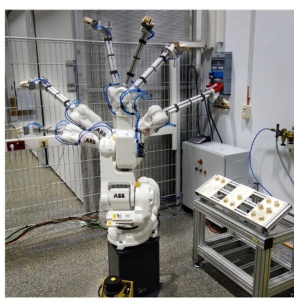


(**d**) Robot placing finished part in different tray.

**Fig. 7.** Example machine tending application.

(**a**) $TM_{front}$

(**b**) $TM_{low}$



(**c**) $TM_{top}$

**Fig. 8.** Illustration of different transfer paths between tray and machine locations.

1. direct path via the front side of the robot ($TM_{front}$)
2. via the top above the robot's base ($TM_{top}$)
3. low path, where the gripper is turned downwards before rotating around joint 1 between the tray / machine locations ($TM_{low}$)

While the path $TM_{front}$ allows the fastest uninterrupted transfer movement, the horizontal space swept by the robot arm is significantly larger, and the robot passes by at a closer distance to the dummy's approach location (which can result in a lower permissible robot velocity).

Table 3 lists the total cycle time, average and maximum velocity for each transfer path when the application is running *without* any operator interaction. In a real machine tending application, this would require some kind of automatic feeding system, which is usually expensive and inflexible, and therefore only pays off for sufficiently large lot sizes. Since our focus is on collaborative applications instead, our experimental trials include the regular interruption of robot motion by the operator entering the robot cell.

**Table 3**
Application cycle time, average and maximum velocity for each transfer path (uninterrupted robot operation).

| Transfer motion | Cycle time $T_c$ [s] | Avg. velocity $\bar{v_r}$ [mm/s] | Max. velocity max ($v_r$) [mm/s] |
|---|---|---|---|
| $TM_{front}$ | 144.1 | 1040 | 5000 |
| $TM_{top}$ | 164.8 | 935 | 3875 |
| $TM_{low}$ | 168.1 | 839 | 3480 |

## 4.3. Reference safeguarding concepts

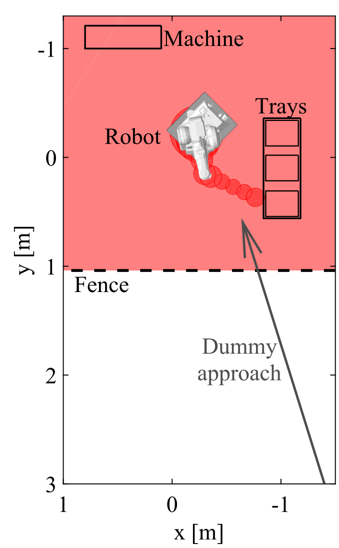
For comparison with both dSSM approaches, two conventional safeguarding methods were realized in the lab setup as references:

First, a robot cell separated by fences with interlocked gates was emulated (Fig. 9 a), in which the robot operates at full speed as long as no person is nearby, but motion is stopped completely once the dummy that acts as the operator reaches a zone just outside the robot's workspace (i.e. enters the cell via the safety door). When the dummy exits the "fenced" area, an additional 5 s delay before the robot operation continues accounts for operator actions that would be necessary in a real application, e.g. closing the safety door, checking that the cell is vacant and acknowledging via a button press.

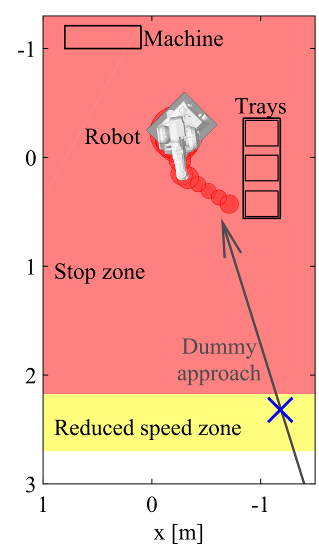
As a second reference concept, a zone supervision by the laser scanner has been implemented (Fig. 9 b), in which violation of an outer "reduced speed zone" triggers a speed reduction to a fixed limit of 1.35 m/s, that is 0.5 m in 0.37 s. This limit results from ergonomics considerations according to Bortot [48], who mentions the values of 0.5 m for a comfortable distance to a hazard and 0.37 s as acceptable lead time for reaction. Furthermore, violation of the inner "stop zone" issues a halt of robot motion. When the inner zone becomes free, robot motion can continue automatically. The inner zone was dimensioned according to ISO 13855:2010 [12].

In summary, the prototype setup allows to compare four different safeguarding principles:

1. Fenced robot cell, where operator access is possible only via a safety




(**a**) Emulation of a fenced robot cell. Robot motion is stopped when the dummy passes the "fence" location. A delay after the red area becomes free accounts for operator actions necessary to continue operation.




(**b**) Zone-based supervision. Upon the dummy entering the yellow zone, the robot speed is reduced to a fixed limit. Violation of the red zone issues a stop. The application continues automatically once the red zone becomes vacant.

**Fig. 9.** Reference safeguarding concepts.

door, which will cause robot motion to stop entirely when opened.
2. Zone-based supervision via the laser scanner, with two fixed zones which trigger a speed reduction or stop of robot motion respectively upon operator approach.
3. *dSSM 1D*, which adjusts robot speed based on the distance to the operator (Section 3.2).
4. *dSSM 2D*, which adjusts robot speed based on the distance to the operator and the direction of robot motion (Section 3.3).

## 5. Experimental evaluation

Several physical trials with the prototype setup were conducted to compare the different safeguarding principles in terms of application productivity. In these trials, a number of application parameters have been varied, which were presumed to have an impact on cycle time. Different test cases with relevance to industrial practice have been defined based on this parameter space, as described in the following.

### 5.1. Parameters & test scenarios

Varying the application parameters, especially those concerning the operator interaction, allows both investigating their individual effect on the application as well as establishing conclusions concerning the suitability of the safeguarding concepts for different types of applications. The parameters which are of particular interest are:

- The degree of collaboration between the operator and the robot, i.e. the fraction of the total application cycle time that the operator is present within the shared workspace:
  - *Intermittent collaboration*: The robot operates independently for the majority of the time; occasionally, the operator needs to enter the robot cell. Thereby, the following parameters of these operator interactions can be adjusted:
    - Frequency at which the operator enters the robot cell (i.e. number of interactions per minute or per application cycle),
    - Point in time at which the operator enters the robot cell (relative to the current state of the application, e.g. at the instant the robot starts the transfer motion towards the machine)
  - *Continuous collaboration*: The operator and the robot share their workspace during most of the application.
- For both intermittent and continuous collaboration, the transfer path that the robot uses to transport the parts between the tray and machine locations can be altered, as presented in Section 4.2, thereby changing the space swept by the manipulator as well as its distance with respect to the dummy's approach location.

Based on these possibilities, the following test scenarios have been defined for the experimental evaluation in our prototype setup:

1. *Intermittent collaboration*
   (a) The operator enters the robot cell with a defined frequency (once per minute, i.e. three times per cycle). The time of operator access relative to the current application state is consistent across all trials.
   (b) The frequency of operator access to the robot cell is varied between trials (one to three times per minute, i.e. three to nine times per cycle). The time of operator access relative to the current application state is kept consistent.
   (c) The operator enters the robot cell with a defined frequency (twice per minute, i.e. six times per cycle). Separate trials are conducted for three different settings of operator access timing relative to the current application state.
2. *Continuous collaboration*: The operator is permanently present at a defined point in the robot cell. This point is chosen such that it is within the *reduced speed zone* in case of the zonal supervision (as indicated by the blue X in Fig. 9 b), otherwise no robot motion would be possible for this safeguarding principle.
3. The transfer path of the robot is altered between the three predefined settings for all of the above test cases.

The first test case represents the *normal operation* of the collaborative machine tending application: The robot feeds material to the machine(s) and extracts processed parts; the operator enters the cell in regular intervals for the exchange of part containers. Test cases 1b and 1c add variation in frequency and timing of these interruptions. This is particularly important in order to gain insight into scalability and transferability of the results to other collaborative applications, which will most likely differ with respect to these parameters.

Test case two represents a different type of application, such as a collaborative assembly task, which benefits from a smaller distance between the worker and the robot (e.g. by reducing time required for parts handover). This case also resembles the commissioning phase of a robot application.

Finally, the third case is intended to investigate the influence of the different robot paths, which differ in their execution time as well as their distance to the operator's location. Tuning the robot path to optimize the application cycle time is a common task in the process of commissioning industrial robot applications. In collaborative settings however, the distance to the operator is another aspect to factor in, which has a direct influence on permissible robot speed in case of SSM.

### 5.2. Productivity metric

The productivity of the machine tending application for the described test scenarios when using a particular safeguarding concept was assessed by the total application cycle time (where one cycle corresponds to the handling of three trays with six parts each). Furthermore, the velocity profile and distribution were examined for certain test cases. We assume that the application requires regular operator intervention. Thus, instead of an uninterrupted application cycle, the cycle time when running the application in a fenced cell that is regularly accessed by the operator served as reference for rating the performance of the fence-less safeguarding concepts.

## 6. Results

The particular cycle times achieved with the different safeguarding principles in each of the test cases as defined in Section 5.1 are presented in the following. The discussion in Section 7 provides further insight into the experiments by examining aspects like robot speed and standstill duration.

### 6.1. Intermittent collaboration (Test case 1)

#### 6.1.1. "Normal operation" (1a)

During the trials for the test case 1a, which represents the *normal operation* of the machine tending cell, the dummy approached the robot once one of the three trays was completely loaded, i.e. three times during one application cycle (approximately once per minute). The resulting total cycle times are given in Table 4 for all safeguarding methods and transfer paths.

#### 6.1.2. Varying the frequency of operator access (1b)

Trials from test case 1a have been repeated with two and three times higher frequency of operator access to the robot cell to investigate the influence of this parameter on application productivity. The resulting cycle times when using the transfer path $TM_{front}$ are presented in Table 5. The experiments with the other two transfer paths $TM_{top}$ and $TM_{low}$ produced comparable results.

#### *6.1.3. Varying the point in time of operator access (1c)*

Another set of experiments varied the point in time at which the operator interrupts robot operation with respect to the current application state between three settings $t_1$ / $t_2$ / $t_3$, while keeping the frequency of operator access consistent at approximately two times per minute (six times per cycle). The measured cycle times are shown in Table 6.

### *6.2. Permanent operator presence (Test case 2)*

In test case 2, the dummy which represents the operator stays in the vicinity of the robot during the entire application. Obviously, this scenario can only reasonably be implemented with the fence-less safeguarding principles. Table 7 presents the obtained cycle duration.

### *6.3. Changing the transfer path (Test case 3)*

Both the test cases for intermittent as well as continuous collaboration between the operator and the robot were repeated with each of the three transfer paths. The results obtained from this comparison, i.e. measured cycle times divided by transfer path, have been presented in Tables 4, 6 and 7.

### *6.4. Summary of results*

In all experimentally investigated scenarios, the dSSM approaches led to a cycle time advantage compared to the conventional safeguarding methods. *In comparison to a fenced robot cell*, the total application cycle time was reduced by

- 4% to 13% when using the dSSM 2D method,
- 3.4% to 11.2% when using dSSM 1D.

Particularly significant reduction in cycle time can be achieved by using dSSM in applications where the operator is present in the vicinity of the robot for longer periods or permanently. In our example task, the cycle time for this test scenario was shortened *compared to a conventional zone monitoring approach* by

- up to 18.1% when using dSSM 2D, and
- up to 11% with dSSM 1D.

## 7. Discussion

The following sections discuss the presented results and take a closer look at certain details of the particular test scenarios. In particular, we analyze the influence of the application parameters that have been varied in the different test cases on the potential performance benefit of dSSM.

### *7.1. Intermittent collaboration (Test case 1)*

#### *7.1.1. "Normal operation" (1a)*

The obtained application cycle times revealed that the fence-less safeguarding concepts bring a clear benefit compared to a fenced robot cell. Thereby, the dSSM approaches outperformed the zone-based supervision; the shortest cycle time was achieved when using the dSSM 2D approach combined with the transfer path $TM_{front}$, which reduced the cycle time by 4.6% compared to a fenced cell.

**Table 4**
Application cycle time for test case 1a (Normal operation).

| Safeguarding concept | Cycle time [s] per transfer path | | |
|---|---|---|---|
| | $TM_{front}$ | $TM_{top}$ | $TM_{low}$ |
| FENCE* | 176.9 (100%) | 197.7 (100%) | 200.3 (100%) |
| ZONES | 173.6 (−1.9%) | 193.0 (−2.4%) | 196.2 (−2.0%) |
| dSSM 1D | 169.4 (−4.2%) | 189.7 (−4.0%) | 193.4 (−3.4%) |
| dSSM 2D | 168.8 (−4.6%) | 188.3 (−4.8%) | 192.3 (−4.0%) |

* The dummy accesses the collaborative workspace 3 times over one full cycle, with a restart delay of 5 s each time (cf. Section 4.3); hence the restart delay accounts for a total of 15 s in the measured cycle time.

**Table 5**
Application cycle time for test case 1b (Varying the frequency of access of the operator) when using the transfer path $TM_{front}$.

| Safeguarding concept | Cycle time [s] for given frequency of operator access | | |
|---|---|---|---|
| | 1 $min^{-1}$ | 2 $min^{-1}$ * | 3 $min^{-1}$ |
| FENCE | 176.9 (100%) | 186.0 (100%) | 207.0 (100%) |
| ZONES | 173.6 (−1.9%) | 179.3 (−3.6%) | 196.6 (−5.0%) |
| dSSM 1D | 169.4 (−4.2%) | 172.6 (−7.2%) | 183.8 (−11.2%) |
| dSSM 2D | 168.8 (−4.6%) | 169.2 (−9.0%) | 182.3 (−11.9%) |

* Shorter dwell time of the operator inside robot cell, compared to the other trials.

**Table 6**
Application cycle time for test case 1c (Varying the point in time of operator access to the robot cell).

| Safeguarding concept | Time of operator access | Cycle time [s] per transfer path | | |
|---|---|---|---|---|
| | | $TM_{front}$ | $TM_{top}$ | $TM_{low}$ |
| FENCE* | $t_1/t_2/t_3$ | 186.0 | 206.1 | 209.8 |
| ZONES* | $t_1/t_2/t_3$ | 179.3 | 197.2 | 200.3 |
| dSSM 1D | $t_1$ | 170.6 | 189.4 | 195.1 |
| | $t_2$ | 177.5 | 191.8 | 194.4 |
| | $t_3$ | 169.6 | 190.6 | 194.3 |
| dSSM 2D | $t_1$ | 161.8 | 181.6 | 190.2 |
| | $t_2$ | 176.7 | 191.8 | 194.2 |
| | $t_3$ | 169.2 | 189.7 | 193.3 |

* Computed / average values provided for comparison. In practice, the timing of operator access has barely any impact on cycle time in case of zone-based safeguarding or a fenced cell, except for slight deviations depending on the current programmed robot speed at the time of operator access, which causes differences in acceleration time after the cell becomes vacant.

**Table 7**
Application cycle time for test case 2 (Permanent operator presence).

| Safeguarding concept | Cycle time [s] | | |
|---|---|---|---|
| | $TM_{front}$ | $TM_{top}$ | $TM_{low}$ |
| ZONES | –* | 238.3 (100%) | 259.3 (100%) |
| dSSM 1D | 238.5 | 216.1 (−9.3%) | 230.9 (−11.0%) |
| dSSM 2D | 207.1 | 200.3 (−15.9%) | 212.3 (−18.1%) |

* The *stop zone* in case of $TM_{front}$ is larger due to the increased area swept by the manipulator. The selected dummy position violated this zone, thus no measurement was possible for this combination.

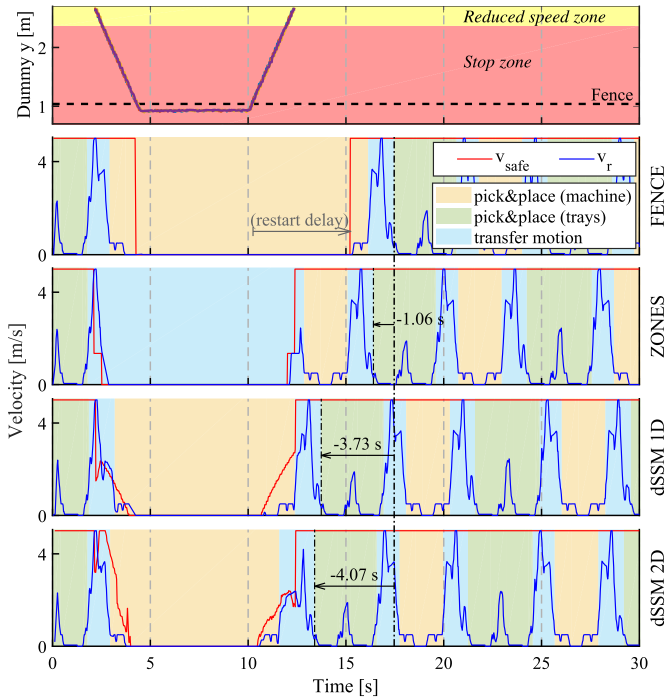
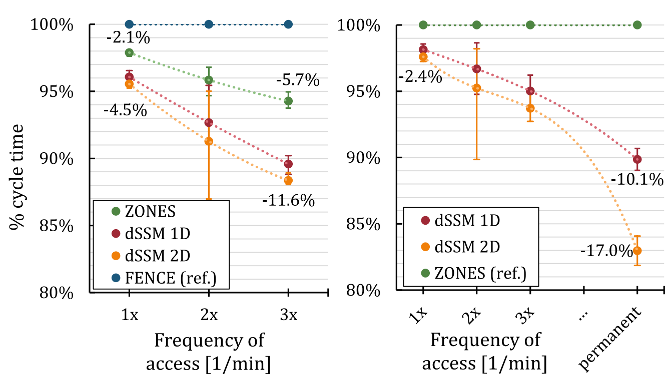



**Fig. 10.** TCP velocity profile and dummy position (top plot) for test case 1a and all safeguarding concepts. The plots show the first time where the dummy entered the robot's vicinity. The robot was using the transfer path $TM_{front}$. The background shading illustrates the different application phases. As indicated, the fence-less safeguarding methods had a lead over the fenced robot cell after the dummy retreated.




**(a)** Zone-based and dSSM compared to fenced safeguarding, **(b)** dSSM compared to zone-based safeguarding.

**Fig. 11.** Cycle time reduction depending on frequency of operator access to the robot cell (error bars indicate full range of observed values, i.e. for all transfer paths, and all points in time of operator access in case of a frequency of 2 $min^{-1}$).

A detail of the recorded velocity profile for all tested safeguarding concepts is depicted in Fig. 10, which shows the first 30 s of each trial, when the dummy entered the vicinity of the robot for the first time. The top plot shows the dummy motion, which was the same in all trials, and indicates the locations of the fence and zone boundaries. The subsequent plots depict the velocity limit $v_{safe}$ that was sent to the robot controller, and the resulting velocity profile of the robot's TCP for each of the evaluated safeguarding concepts. The application has been divided into three application phases, which are indicated by the background shading. The following observations can be made based on these diagrams:

- Taking as a reference the time when the robot reached the trays for the second time (marked by the 2nd green shaded area), the zone-based and dSSM concepts were a certain time ahead compared to the fenced safeguarding after the dummy left, as indicated in the respective plots. Although with the fenced cell a stop of robot motion was issued at the latest point in time, the 5 s restart delay led to the longest standstill time. Since these time lags add up when the dummy approaches the robot multiple times, a significantly shorter total cycle time resulted for the dSSM approaches compared to the other concepts.
- The dummy started approaching the robot during a transfer motion. Since the robot was moving at high speed during this phase, there is a clear difference in how the speed limit affected robot motion between the four trials. It could be expected though, that the difference between the safeguarding concepts is less significant if the dummy would move towards the robot in a different application phase, where the robot is programmed to move at lower TCP velocity. This has been investigated in test case 1c, which is discussed in Section 7.1.3 below.

### 7.1.2. Impact of the frequency of operator access (1b & 2)

The results from test scenario 1b indicate that the cycle time advantage through use of fence-less safeguarding concepts in comparison to a fenced robot cell increases with the frequency of operator access to the robot during the course of the application.

The diagrams in Fig. 11 show the cycle time as a percentage with reference to the fenced or zone-based safeguarding concept depending on the access frequency. In the present example application, a reduction in cycle time of 5.7% could be achieved by using zone-based supervision instead of fencing and interlocked gates, whereas the potential reduction with dSSM 2D was around twice as much.

Fig. 11 b includes the case of permanent operator presence near the robot (test case 2), where the dSSM methods led to a significantly shorter cycle time in comparison to a conventional zone supervision; it is furthermore interesting how the potential cycle time savings increase when considering the direction of robot motion in dSSM 2D in contrast to the one-dimensional approach. Further details of this scenario are discussed in Section 7.2.

### 7.1.3. Impact of the timing of operator interaction (1c)

By varying the time at which the dummy starts to move towards the robot between three settings $t_1$ to $t_3$, the earlier assumption was confirmed: the time instant at which an operator approaches the robot – more specifically the particular conditions such as robot position and programmed speed at that moment – affects the impact of this intervention on the application performance.

A comparison of the resulting velocity profiles when using the dSSM 2D approach is presented in Fig. 12. Note that the dummy motion (velocity, front position and dwell time) was the same in all three trials, and shifted in time only. Nevertheless, analysis of the robot velocity shows a clear difference in the standstill time of the robot, as indicated in the plots. The reason for this becomes obvious when studying the current robot motion at the time of the intervention – Fig. 13 illustrates the last portion of the TCP path before stopping and the position at which the robot came to a halt for all three cases:

In trial $t_1$, the robot was just moving away from the approaching dummy, which caused the permissible robot velocity $v_{safe}$ to increase shortly after the first dip. The manipulator stopped on the machine side, at a distant location with respect to the dummy position. In contrast, in both trials $t_2$ and $t_3$, the robot was about to move in the direction of the dummy, and stopped at a much closer position. Consequently, it needed to stop earlier, and had to wait longer for the dummy to retreat to a

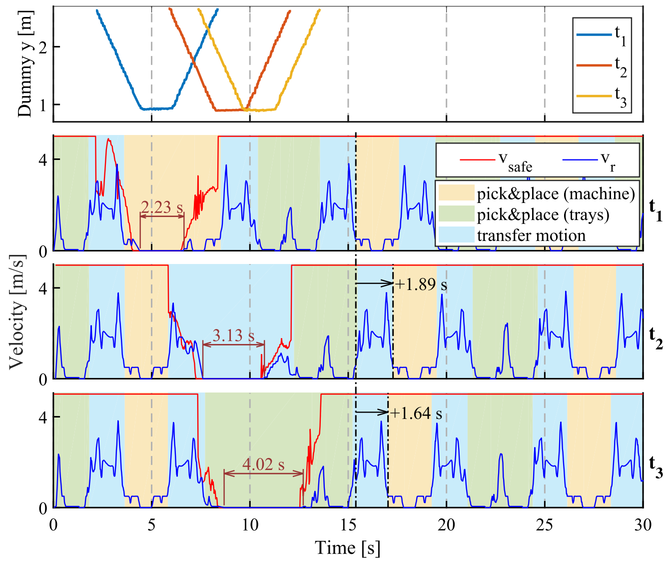



**Fig. 12.** TCP velocity profile and dummy position for different points in time at which the dummy approached the robot (test case 1b), using dSSM 2D. The robot was using transfer path $TM_{top}$. Indicated is the standstill duration for each case, as well as the time lag compared to trial $t_1$ after the dummy has withdrawn.

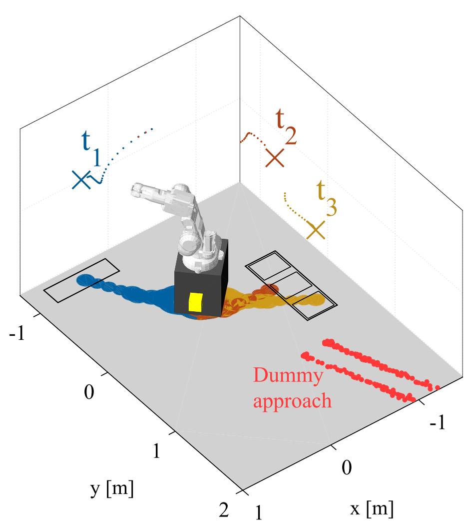



**Fig. 13.** Stop points of the robot when the dummy approached at different points in time (test case 1b), using dSSM 2D and $TM_{top}$.

more distant position compared to $t_1$, which resulted in an increased standstill duration.

It is furthermore worth noting that, even though trial $t_3$ shows the longest standstill period, the time lag after the dummy retreated was slightly smaller compared to trial $t_2$ – see the indicated differences in reaching the application phase *pick & place (machine)*. This can be attributed to the short periods of speed reduction before and after standstill of the robot: In $t_2$, the robot was still in the *transfer motion* phase (where the programmed velocity is large) when the dummy advanced, hence the necessary speed reduction had a significant impact on the traveled distance. In trial $t_3$ by contrast, the robot had just finished the transfer motion and entered the *pick & place* phase (with lower programmed speed) before it stopped, such that the application progress was affected less by the speed reduction phases. This shows that the *standstill time* alone is not a representative measure to draw conclusions on the overall performance. Note that the impact of the timing while using zone-based safeguarding or dSSM 1D was small compared to the influence on cycle time with dSSM 2D, which evidences the particular importance of the robot's movement direction at the time of operator access.

In practice, this effect could be accounted for by training of the operator, as well as by providing a visual or audible signal, indicating a suitable moment to enter the robot cell with minimal impact on the productivity of the running application.

### *7.2. Permanent operator presence (Test case 2)*

The application cycle times measured for the *continuous collaboration* scenario, in which the operator was permanently at a fixed position near the robot's workspace (at a position in the reduced speed zone in case of zone-based safeguarding), show considerable benefits of the dSSM approaches over the zonal safeguarding. Using the transfer path $TM_{low}$, a reduction in cycle time of 18.1% was observed when using dSSM 2D instead of zone supervision. This significant performance gain can be attributed to the increase in average robot speed that is enabled by the dynamic limitation when using dSSM, in comparison to the fixed speed limit that is applied in the zones setup.

A detail of the recorded robot's TCP velocity for the different safeguarding methods is presented in Fig. 14. The top plot shows the constant speed limit $v_{slow} = 1.35\,m/s$ that is applied to the robot motion in case of violation of the reduced speed zone. With the dSSM 1D concept in contrast, the permissible velocity varied between approximately 0.8 and 2 m/s, depending on whether the robot is currently turned towards the direction of the dummy, or to the opposite side. The largest permissible robot velocity resulted when the dSSM 2D approach was used, as shown in the third plot. In this case, the TCP reached a maximum momentary velocity of above 3.8 m/s, while the robot was turning away from the dummy location. The individual time advantages of the dSSM approaches after handling of one part as compared to the zone concept are indicated in the plots.

The diagrams in Fig. 15 show the distribution of TCP velocity values observed during one application cycle for each of the safeguarding methods, divided by application phase; the dashed lines indicate the mean velocity in each phase.

The frequency of velocities observed in the pick & place phases are very similar in all three trials; their associated mean values are nearly identical. The programmed robot velocities of 50 mm/s and 500 mm/s in these phases are clearly recognizable. Notable differences are found, however, during the phases of transfer motion. While the robot moved

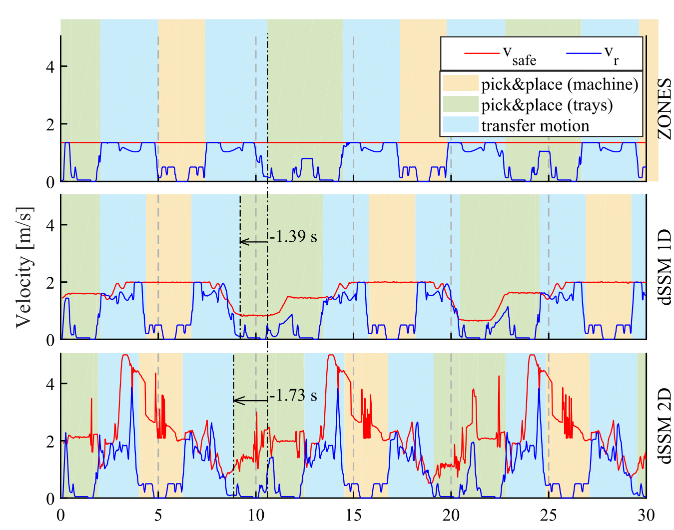



**Fig. 14.** TCP velocity profile for zone-based and dSSM safeguarding recorded during permanent presence of the dummy in the robot's vicinity (test case 2), using transfer path $TM_{top}$.

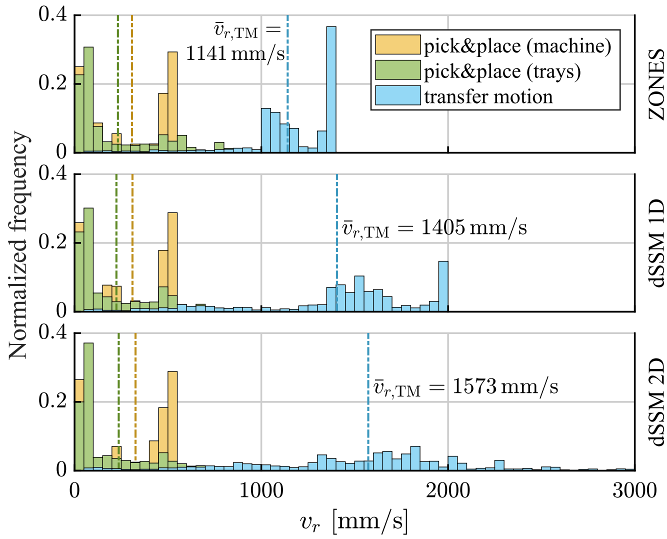



**Fig. 15.** TCP velocity distribution for permanent presence of the operator in the robot's vicinity (test case 2), using transfer path $TM_{top}$. The dashed lines indicate the average velocity of each application phase. Each bin comprises observed velocities within an interval of 50 mm/s width.

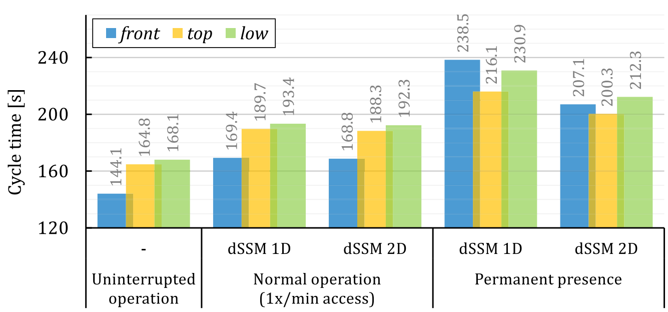



**Fig. 16.** Cycle time comparison for different transfer paths.

at $v_{slow}$ for the major part in the ZONES measurement, the TCP velocity during transfer motion tended towards larger values for the dSSM approaches, which is reflected in a larger mean velocity in these trials. This examination confirms that dSSM allows a performance gain particularly during transfer motion, whereas other application phases, in which the robot is programmed to move at a lower speed, are of minor importance.

### *7.3. Influence of robot path & distance to probable operator locations (Test case 3)*

In many cases, the typical approach direction of the operator is known in advance with sufficient accuracy to be be accounted for when programming the robot path. We repeated all physical trials with each of the three different paths as introduced in Section 4.2. Examination of the recorded application cycle times revealed that the increased distance to the operator location by using the transfer paths $TM_{top}$ or $TM_{low}$ and the associated higher permissible robot speed did not have a sufficiently strong impact to allow recouping of the initial handicap during most experiments. However, in the case of permanent operator presence in the vicinity of the robot, the more severe speed reduction required with $TM_{front}$ due to the closer distance to the dummy location led to a longer total cycle time compared to the application running with the transfer path $TM_{top}$. This effect was even more prominent when using the dSSM 1D method, in which case both the trials using $TM_{top}$ as well as $TM_{low}$ finished faster than the one where $TM_{front}$ was selected. Fig. 16 compares the application cycle time that was achieved with each transfer path for uninterrupted robot operation as well as using the dSSM approaches for test cases 1a and 2.

It can be concluded that an optimization of the robot path with respect to the distance to probable operator locations should be considered to improve application efficiency whenever the operator is expected to sojourn near the robot for extended periods. In addition, an active modification of the robot path at run-time could be considered – in the simplest form e.g. by switching between several programmed paths depending on the operator position.

### *7.4. Suggestions for further work*

While the present work has suggested two possible approaches for the dynamic adaptation of robot velocity in collaborative applications using SSM, there are numerous detail questions to be resolved before dSSM is ready for large-scale industrial deployment.

#### *7.4.1. Applicability of results to other application types*

The present study is based on a specific application, and while we expect that other applications will show somewhat different behavior in detail, they will most likely be influenced by the same factors as studied in this paper. In our view, machine tending is a particularly interesting application, since the alternative to manual tending of material trays, i.e. automatic material feed-in/feed-out, is considerably more expensive in initial investment as well as in upkeep and changeover.

#### *7.4.2. Methodology and tools for assessing productivity of SSM applications*

In the present work, the productivity of a machine tending task was gauged by the total application cycle time. This provides a reasonable comparative metric for our example task, though no quantitative conclusions related to other SSM applications and implementations can be drawn. Generally, the cycle time of collaborative applications cannot be defined in a strict manner, but rather represents a temporal average over variations due to irregularities in the behavior of the operator. The challenge is thus to define generalized metrics and/or specific reference application cycles to compare different SSM implementations as well as different modes of collaborative robot operation.

We suggest that simulation-based assessment of collaborative applications, targeting the suitability of risk reduction and the expected productivity, can be an effective way forward. Various safeguarding options can then be compared before investing in physical installations. This reduces the investment uncertainty of the end customer when considering a collaborative application and can support the risk assessment process in detail once a safeguarding variant has been chosen. In such simulations, the tasks of the operator must be modeled in sufficient detail to enable the analysis of productivity. To assess also the safety of the application, realistic perturbations on the behavior of the operator must be modeled so as to verify the proper response of the safety concept.

With a model of the SSM application in a simulation environment, another question touching on the economics of a projected application can be more easily addressed, namely that of the expected floor space requirement for an envisioned installation. Also, when collaborative applications are intended to be set up in an existing factory environment, the available space can be a hard constraint that must be met. The floor space needed is influenced by the application layout and use cases, by the details of robot and operator motion, by the limitations of sensor systems and SSM supervision algorithms, and by the stopping behavior of the robot. In this paper, we have focused on the influence of different supervision principles on the cycle time, given a fixed spatial arrangement, varying the behavior of the operator.

#### *7.4.3. Sensory workspace supervision*

As noted, the workspace sensing used here was not capable of returning the position of the operator in a safety-rated manner. Sensor products with this ability are, however, a prerequisite for deployment of dSSM solutions that conform to the requirements of ISO 10218-2

[21]. In addition, the integration of a future safety-rated sensing system into an SSM application must also address the reliability of the calibration of the sensor data and the possible need for periodic testing.

Given a viable system for workspace supervision, the SSM method can gain from improved modeling and prediction of operator motion. While the requirements of ISO 13855 [12] force us to assume certain worst case operator behavior, these assumptions are meant to cover for the lack of more detailed information on operator motion. More realistic modeling and prediction has the potential to recover additional productivity in SSM applications without compromising safety.

Another challenge is the possibility that inanimate objects not expected in the workspace can lead to a throttling or standstill of the application, even without presence of an operator. Today, such situations must be managed by work procedure discipline, but additional flexibility might be gained if the sensing system were able to discriminate reliably between animate and inanimate objects in its field of view.

#### 7.4.4. Behavior of robot

In general, the stopping behavior of an articulated robot is more complex than the simplified model assumed in Eq. (3) above. Typically, the control algorithms applied, the mechanical braking behavior, and the robot motion state interact in a complex manner.

Referring to Eq. (2), we recall that the stopping distance of the robot $S_s$ figures centrally in the formulation of the protective separation distance for SSM. In fact, the stopping time is also implicitly contained in the distance $S_h$ prospectively covered by the operator during the system response latency and the stopping motion.

Thus, a proper implementation of the SSM supervision criterion in Eq. (1) must include a safety-rated computation of the anticipated stopping behavior of the robot, based on its momentary pose, velocity and payload. Suitable algorithms should execute in a real-time manner on the time scale of the refresh rate of the workspace supervision.

While this work has focused decidedly on adapting the robot velocity on the given path in the application, it is possible to satisfy the supervision criterion in Eq. (1) by adapting the path itself. This can take the form of a run-time selection among a fixed number of path alternatives, choosing the one that maximizes the allowable robot trajectory speed. A more general approach would comprise the dynamic generation of the robot trajectory to maximize the distance to the operator and the allowable robot velocity, again in a safety-rated implementation.

When using manipulators with one or more degrees of kinematic redundancy, it is conceivable that a null-space pose optimization criterion can be derived from the requirement to maximize the distance between the operator and the manipulator.

## 8. Conclusions

We developed and validated the performance of two approaches for the continuous adaptation of robot speed in a SSM-type collaborative robot application. The experiments indicated a clear benefit of the developed methods in terms of a reduced application cycle time for all investigated test scenarios in comparison to conventional safeguarding. The trials furthermore revealed the impact of several factors that should be considered during application design. Among these are the frequency, timing and duration of operator presence near the robot, the velocity profile of the application, and the robot path and its distance to expected operator locations. Since most of these factors are highly application specific, the use of suitable simulation-based tools to assess the potential economic benefit and select the optimal solution for a particular application seems advisable.

The application of the developed methods in industry will depend on advances in the field of safety sensing, in particular on the safe communication of operator locations. Future work in this domain could address topics such as reliable tracking and prediction of operator motion, and capabilities to distinguish between humans and inanimate objects. Next steps on the side of robot manufacturers might focus on the reliable estimation of stopping distances and providing aforementioned simulation tools in order to support system integrators in the design of efficient collaborative applications.

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